Chemistry Review

1. Complete the following table.

Symbol or Ion	Mass number	Atomic Number	Number of protons	Number of electrons	Number of neutrons
Te	128	52			
V	51		23		
Ca^{2+}	40		20		
S^{2-}		16			16

2. a) Draw electron configuration diagrams for chlorine, aluminium and magnesium.
   b) Write the electron configurations for those elements given in part a.
   c) Which of the elements is most likely to gain electrons if ionically bonded?
   d) Which element forms an ion with the same electron configuration as neon?

3. Write correct formulae for each of the following substances.

   a) Zinc hydrogencarbonate
   b) Manganese hydroxide
   c) Aluminium sulfate
   d) Lithium phosphate
   e) Hydrochloric acid

4. Name the following substances.

A) HNO_3
B) CO_2
C) CuO
D) $MgCO_3$

5. Balance the following equations.

   a) $N_2 + H_2 \rightarrow NH_3$

   c) $Fe_2O_3 + CO \rightarrow Fe + CO_2$

   e) $Cu + HNO_3 \rightarrow Cu(NO_3)_2 + NO + H_2O$